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Evaluation of Environmental Parameters on Remediation Efficiency: Mathematical Modeling for Coliform Removal

Robert León-Suárez[1]

***Abstract* — Phytoremediation using aquatic macrophytes is a sustainable strategy for the removal of coliforms and *Escherichia coli* in wastewater, although its efficiency is subject to variations in environmental parameters such as pH, temperature, and total dissolved solids (TDS). This study developed a first-order kinetic mathematical model, incorporating an environmental penalty mechanism based on a Gaussian function to dynamically adjust the removal coefficient (k) according to physicochemical conditions. Six species of macrophytes (*Azolla*, *Salvinia*, *Pistia*, *Ceratopteris*, *Spirodela*, and *Eichhornia*) were experimentally evaluated alongside a positive control (Lemna) and a negative control (no plant), with daily measurements of bacterial concentrations and environmental parameters over seven days. The results showed that the model adequately described removal trends in several treatments, especially for *E. coli*. The environmental penalty allowed pH, temperature, and TDS effects on k to be evaluated, producing more conservative predictions. *Ceratopteris* and *Azolla* showed the best fits for coliforms ($R^2 > 0.99$), while *Spirodela*, *Salvinia*, and *Eichhornia* exhibited low performance. It is concluded that the model is a useful tool for predicting removal efficiency under variable environmental conditions, though further research into species-specific thresholds and microbial variables is required to optimize its applicability in real-world scenarios.**

Keywords: phytoremediation,* Escherichia coli*, Coliforms, mathematical modeling, environmental penalty, aquatic macrophytes.

***Resumen* — La fitorremediación con macrófitas acuáticas se presenta como una estrategia sostenible para la eliminación de coliformes y *Escherichia coli* en aguas residuales, aunque su eficiencia está sujeta a variaciones en parámetros ambientales como pH, temperatura y sólidos disueltos totales (TSD). Este estudio desarrolló un modelo matemático de cinética de primer orden, incorporando un mecanismo de penalización ambiental basado en una función gaussiana para ajustar dinámicamente el coeficiente de remoción (k) según las condiciones fisicoquímicas. Se evaluaron experimentalmente seis especies de macrófitas (*Azolla*, *Salvinia*, *Pistia*, *Ceratopteris*, *Spirodela* y *Eichhornia*) junto con un control positivo (Lemna) y uno negativo (sin planta), midiendo diariamente las concentraciones bacterianas y los parámetros ambientales durante siete días. Los resultados mostraron que el modelo describió adecuadamente las tendencias de remoción en varios tratamientos, especialmente para *E. coli*. La penalización ambiental permitió evaluar el efecto del pH, la temperatura y los TSD sobre k, generando predicciones más conservadoras. *Ceratopteris* y *Azolla* mostraron los mejores ajustes para coliformes ($R^2 > 0,99$), mientras que *Spirodela*, *Salvinia* y *Eichhornia* presentaron bajo desempeño. Se concluye que el modelo constituye una herramienta para predecir la eficiencia de remoción bajo condiciones ambientales variables, aunque requiere profundizar en umbrales específicos por especie y variables microbianas para optimizar su aplicabilidad en escenarios reales.**

Palabras Clave: fitorremediación,* Escherichia Coli*, Coliformes, modelado matemático, penalización ambiental, macrófitas acuáticas.

I. INTRODUCTION

BIOTECHNOLOGICAL approaches that employ botanical species such as macrophytes for the mitigation of environmental pollutants are known as phytoremediation [1]. This concept is based on the mechanism of action of certain plants, which includes phytoextraction, phytodegradation, and rhizofiltration, enabling the detoxification of a diverse category of pollutants through physical, chemical, and biological processes [2], [3]. Macrophyte species such as *Azolla caroliniana*, *Eichhornia crassipes*, and *Lemna minor* stand out for their rapid growth and diverse pollutant removal mechanisms [4], [5]. Recent quantitative assessments by [6] and [5] demonstrated that these species can achieve *E. coli* reduction efficiencies exceeding 97.8% and 99.2%, respectively, within controlled environments. Moreover, [7] reported significant removals of organic loads, with ammonia and total suspended solids decreasing by up to 84.4% and 83.2%, respectively, using *E. crassipes*. These results are often attributed to the secretion of antimicrobial flavonoids and the promotion of microbial interactions in the rhizosphere [4], [6]. However, a critical limitation remains these high efficiencies are highly sensitive to biomass saturation and fluctuating wastewater characteristics, such as pH and TDS [7]. Most existing studies offer static performance data but lack kinetic frameworks capable of predicting these non-linear behaviors in field-scale systems [4], [6].

Recent evaluations emphasize that the remediating effect of these species is highly dependent on morphological traits—such as root depth—which facilitate radial oxygen loss (ROL) and promote the microbial interactions necessary for pollutant

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degradation [7], [8]. Furthermore, the success of these biological systems is governed by a complex interplay of environmental factors including nutrient availability, moisture, and electron acceptors which directly influence the metabolic potential and bioremediation activity of the associated microbial communities [9].

Water resources contaminated by microbial pathogens such as coliforms and *E. coli* pose a challenge to global public health, especially in developing countries [10], [11], [12]. Approximately 2.5 million children under the age of five die annually from diarrheal diseases, primarily caused by fecal contamination, particularly from *E. coli* [13]. In Ecuador, a correlation has been detected between the incidence of gastrointestinal diseases and coliform concentrations exceeding permissible limits (greater than 5214 MPN/100 mL) in highly important rivers such as the Yaguachi [11], [13].

Recent studies in the Manabí region [14] have demonstrated the viability of using floating islands with *P. stratiotes* for domestic wastewater treatment, achieving significant removal efficiencies of 80.3% in total suspended solids and 72.1% in turbidity. While these findings underscore the potential of macrophytes in local oxidation lagoons, the research also identified that performance is heavily constrained by seasonal fluctuations in physicochemical parameters, particularly temperature and pH [15]. Furthermore, these studies remain largely descriptive and lack a predictive mathematical framework to quantify how such environmental variations impact the removal kinetics, a gap that the present work aims to address.

This is a critical problem, particularly in rural communities without access to wastewater treatment systems [12], [16]. Despite the potential of phytoremediation for treating water contaminated with coliforms, its efficiency depends heavily on the physicochemical conditions of its development, such as pH, temperature, and dissolved solids concentration [17], [18]. Currently, there is no mathematical model that consistently predicts the kinetics of coliform removal under these conditions, which limits the optimization and scalability of the systems [19], [20].

Historically, design models for these systems have ranged from simple rules to first-order equations; however, these approaches often suffer from high parameter uncertainty and fail to capture the complex dynamic behaviors observed in actual treatment units [21]. In tropical contexts, the kinetic model developed by [9] represents a significant advancement by integrating temperature, sedimentation, and solar radiation into an overall removal rate coefficient ($k = 1.632$ day^{-1}). Their research, conducted in *Typha angustifolia*-based systems, demonstrated that while temperature-induced mortality contributes a rate 0.106 day^{-1} solar radiation is more decisive, contributing 0.356 day^{-1}, ultimately achieving fecal coliform reductions exceeding 99% [9]. Despite these contributions, such deterministic models are highly sensitive to fixed parameters and lack the capacity to account for the stochastic variability of environmental factors like pH and TDS, which introduce significant non-linearities in actual treatment units.

Experimental observations have shown that even in controlled environments, correlations between dissolved oxygen and contact time are often obscured by external environmental factors, making the behavior of the root-zone matrix difficult to predict with simple linear approaches [15].

This limitation is exacerbated by the stochastic nature of natural environments, where fluctuating environmental factors introduce variability not captured by traditional deterministic models [19], [22]. In this context, stochastic frameworks have proven essential for quantifying the partitioning of contaminants, as they account for the probabilistic nature of environmental 'forcing' that determines the ultimate success of the remediation process [19].

Therefore, it is important to contextualize the biological processes involved in the removal of bacterial pollutants through phytoremediation by means of innovative environmental biotechnological tools, such as the application of increasingly diverse mathematical models that integrate key analytical approaches (pH, temperature, and total solids concentration, TDS) to optimize contaminant removal efficiency [23], [24]. Furthermore, from a scientific perspective, the combination of experimental data and computational modeling improves the design of removal systems and allows for adaptation to different geographical contexts, broadening their applicability [25], [26].

This study develops a mathematical model that describes the kinetics of coliform elimination, incorporating the randomness of variables (pH, temperature, and TDS) through an ordinary differential equation, and follows recent advancements in mechanistic state-space modeling, which allow for the simulation of complex phytoremediation dynamics by accounting for the inhibitory effects of fluctuating wastewater characteristics [7].

This approach aims to address the gaps in simple models by including a dynamic environmental penalty coefficient, adjusting the removal efficiency according to changes in physicochemical parameters [23], [27]. The work seeks to evaluate different models using experimental data from 7 plant species and a control, employing statistical metrics such as R^2 and mean squared error [28], [29].

II. METHODOLOGY

A. Material Plant and Wastewater

This study evaluated the phytoremediation capacity of six macrophyte species (*Azolla* sp., *Salvinia auriculata*, *Pistia stratiotes*, *Ceratopteris thalictroides*, *Spirodela* sp., and *Eichhornia crassipes*) in wastewater contaminated with coliforms. Plant material was collected during three field trips to the banks of the Guayas River (Guayaquil-Guayas) and the "Aguas" precinct (Ventanas - Los Ríos). These locations are characterized by a warm tropical climate, with average temperatures between 22–26 °C and a defined rainy season from December to May, conditions that favor the development of aquatic macrophytes. The experimental sampling and bioassays were conducted on June 24, 2016, coinciding with the beginning of the dry season in the region. The selected plants were transported under controlled conditions to preserve their physiological viability until introduction into the bioassays.

Wastewater was obtained from a variable-flow ditch located on the Mapasingue campus of the Universidad Estatal de Guayaquil. Sampling was conducted at four georeferenced points (UTM coordinates, zone 17S: M1 620238 – 9762713; M2 620221 – 9762676; M3 620210 – 9762652; M4 620203 – 9762635), covering areas with and without current to capture spatial heterogeneity in contaminant distribution. The samples were homogenized for four hours under constant agitation to ensure contaminant uniformity before use in the experiments.

B. Acclimation and Experimental Design

Prior to the assays, the macrophytes underwent an acclimation period in containers with water fertilized with Multiflor®, maintained at temperatures between 25–30 °C under natural illumination. Bioassays were conducted in polyethylene experimental units over a period of seven days. The design included six macrophyte species in triplicate, as well as positive (*Lemna minor*) and negative (water without plants) controls. The physicochemical parameters pH, temperature, and (TDS) were recorded daily using a calibrated BIOBASE® PH-900P multiparameter portable meter. The instrument provided a pH resolution of 0.001 pH with an accuracy of ±0.002 pH, and TDS measurements with an accuracy of ±1% F.S., using a default TDS factor of 0.5. Automatic temperature compensation (0–100°C) was employed to ensure data consistency across daily fluctuations.

C. Physicochemical and Microbiological Analyses

Microbiological analyses were performed following the protocol established by ISO standard 9308-1. Concentrations of total coliforms and *E. coli* (CFU/100 mL) were quantified by pour plate method using Chromocult® agar (Merck®). Confirmation of *E. coli* was carried out through Gram stain, oxidase, and Kovacs' indole biochemical tests, ensuring identification specificity.

D. Mathematical Modeling and Calculation of Kinetic Parameters

The mathematical modeling of removal kinetics was based on a first-order exponential decay approach. The removal coefficient *k* was calculated for each day and treatment using the expression (1):

$$k = \frac{ln\left(\frac{C_{t-1}}{C_t}\right)}{\Delta t} \tag{1}$$

Where:
C_{t-1} is the concentration on the previous day,
C_t is the concentration on the current day,
Δt is the difference between the subsequent and previous day (time interval between measurements).

To incorporate the experimental uncertainty inherent to biological processes, a random variability of ±5% was introduced to the *k* values. Extrapolation of the coefficient for day 7 was performed using a second-degree polynomial fit applied directly to the values from days 3 to 6, thus avoiding distortions.

E. Environmental Penalty Mechanism

To evaluate the effect of environmental conditions on system efficiency, a penalty mechanism based on a Gaussian decay function was implemented. The optimal ranges for the model were established using pooled data from all experimental treatments (seven macrophyte species) and the control, including all respective replicates. These ranges were defined as the 95% confidence intervals calculated from the physicochemical parameters recorded during the first two days of the removal process: pH: 7.63–8.06; TDS: 425.59–515.65 mg/L; and temperature: 22.62–24.12 °C. Although this criterion is conservative, it provides a statistically grounded baseline that allows the model to be penalized for extrinsic and variable environmental conditions often encountered in wild systems. The penalty for each parameter was calculated individually using expression (2):

$$f\left(x\right) = \exp\left(-0.5\left(\frac{|x-x_{\text{lim}}|}{\sigma}\right)^2\right) \tag{2}$$

Where *x* is the measured value, x_{lim} is the nearest limit of the optimal range, and σ is the standard deviation of the data from the first two days. The total penalty factor was obtained as the product of the individual penalties for pH, TDS, and temperature. This was applied as a multiplicative factor to the coefficient *k* before calculating the modeled concentrations using the exponential decay equation (3):

$$C_t = C_{t-1} \cdot \mathrm{e}^{-k \cdot \Delta t} \tag{3}$$

Where:
C_t is the modeled concentration on day t,
C_{t-1} is the modeled concentration on the previous day,
k is the removal coefficient (potentially penalized),
Δt is the time interval (1 day).

F. Statistical Validation of the Model

Model validation was performed by calculating accuracy and goodness-of-fit metrics. The mean squared error (MSE), mean absolute error (MAE), and coefficient of determination (R^2) were calculated by comparing the experimental bacterial concentrations with the concentrations predicted by the non-penalized and penalized models, considering exclusively days 2 to 7 to avoid bias from initial conditions. All calculations incorporated robust handling of outliers and missing data by excluding unavailable values. This approach allowed quantification of both the error magnitude and the proportion of variance explained by the model under variable conditions.

III. RESULTS

Comparisons between experimental data and modeled values for the removal of coliforms and *E. coli* through phytoremediation systems using macrophytes are presented. The developed mathematical model, based on first-order differential equation kinetics adjusted through polynomial regression and environmental penalty, demonstrated significant predictive capability under diverse experimental conditions.

Model validation was performed by calculating statistical metrics of accuracy and goodness-of-fit. Mean Squared Error (MSE) and Mean Absolute Error (MAE) values quantified the magnitude of deviations between observed and predicted values, while the coefficient of determination (R^2) assessed the proportion of variance explained by the model.

A. Modeling Without Penalty

Figure 1 compares the experimental and modeled concentrations of total coliforms for eight macrophyte species over the seven-day trial. In the experimental data, five species (*Azolla, Ceratopteris, Eichhornia, Lemna*, and *Salvinia*) reached concentrations ≤200 CFU/100 mL by day 7. While most showed a constant removal trend until day 6, *Eichhornia, Lemna, Pistia*, and *Salvinia* exhibited their highest removal rate on the final day; however, *Pistia* showed slight growth until day 6 before demonstrating a removal effect. In contrast, *Spirodela* presented the lowest removal efficiency during this period.

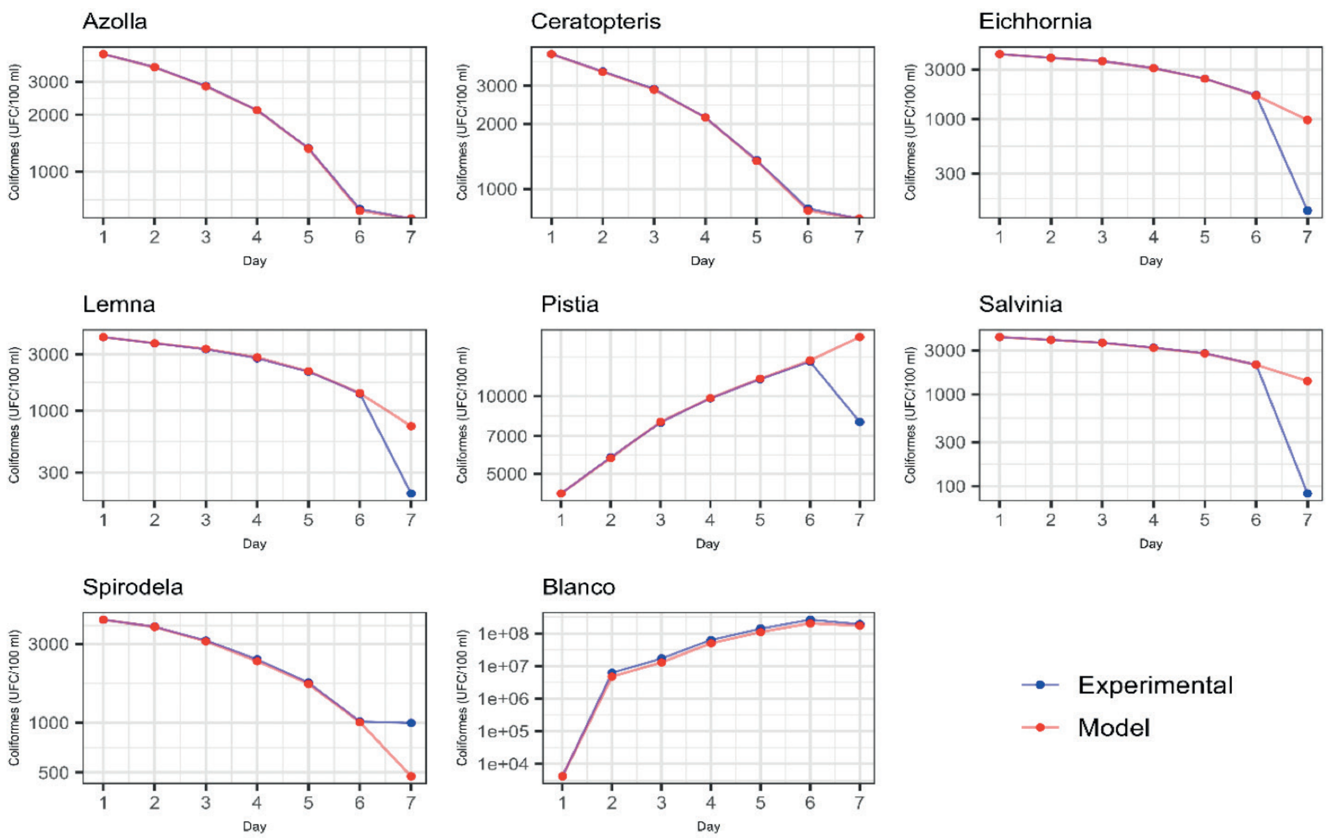

Fig. 1. Experimental versus Modeled Removal Coefficients (*k*) for Total Coliforms.
Note: Error bars representing standard deviation (n=3) are present but appear imperceptible in the graphical representation due to the high magnitude of the initial concentrations and the scale of the Y-axis relative to the low operational variance observed among replicates.

Regarding the predictions, *Azolla, Ceratopteris*, and the control (Blank) showed the highest concordance with the experimental data. For the remaining species, the model failed to fully capture the abrupt variations in removal rates observed experimentally towards day 7, particularly in cases of accelerated or persistently slow removal.

Figure 2 presents the evolution of *E. coli* concentrations in the phytoremediation systems with the eight macrophyte species, showing both experimental and modeled values. The initial concentration was 1200 CFU/100 mL in all cases. Experimentally, all macrophytes achieved complete removal (0 CFU/100 mL) by day 7, while the control maintained detectable concentrations throughout the trial. Unlike the pattern observed for total coliforms, the highest removal rate for *E. coli* occurred during the first day in all plant treatments.

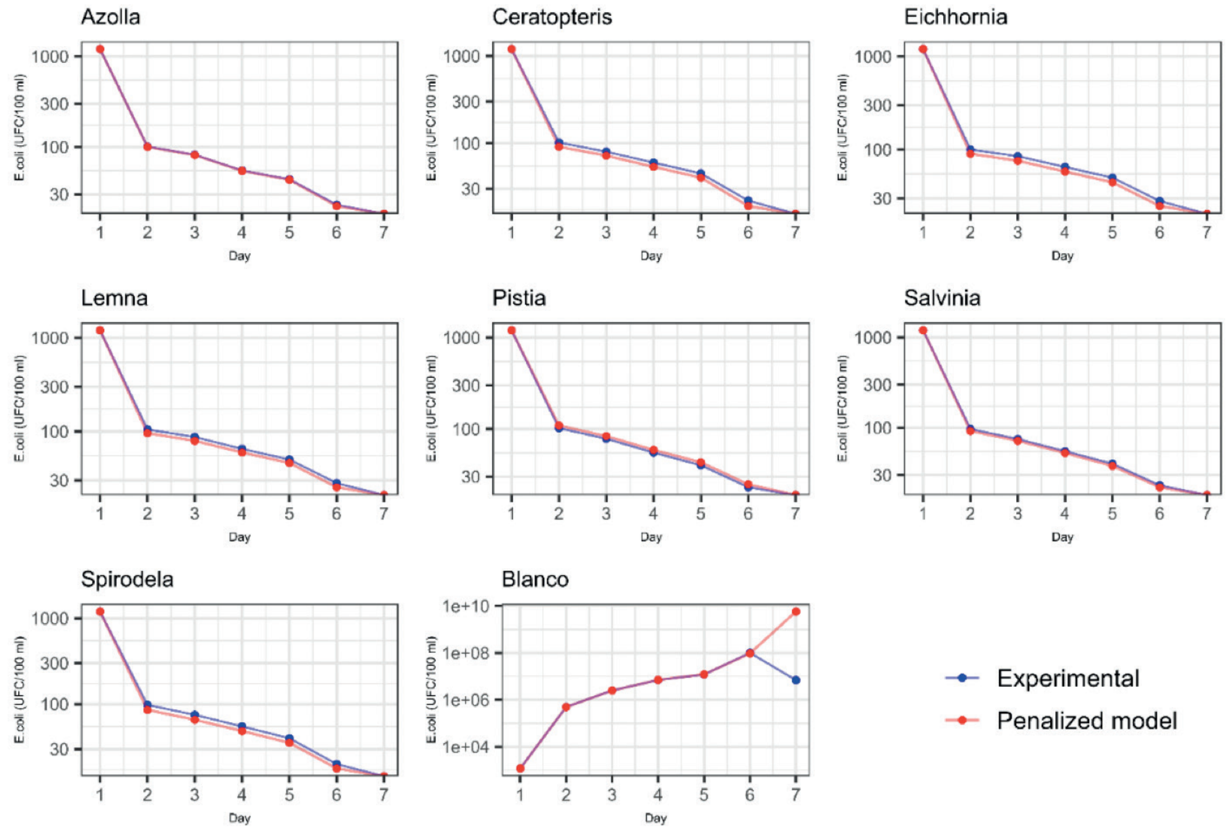

Fig. 2. Experimental versus Modeled Removal Coefficients (*k*) for *E. coli*.
Note: Error bars representing standard deviation (n=3) are present but appear imperceptible in the graphical representation due to the high magnitude of the initial concentrations and the scale of the Y-axis relative to the low operational variance observed among replicates.

The model reproduced the removal kinetics for all macrophyte species with close agreement to the experimental data. The main discrepancy was observed in the control, where the model predicted continuous growth of *E. coli* that did not materialize experimentally towards the end of the trial, a period during which the concentrations showed slight decay.

The validation results for the model without penalty, using three statistical metrics (MSE, MAE, and R^2) for total coliforms and *E. coli*, are shown in table I.

TABLE I
STATISTICAL VALIDATION OF NON-PENALIZED MODELING (COLIFORMS AND *E. COLI*).

Plants	Coliforms			*E. coli*		
	MSE	MAE	R2	MSE	MAE	R2
Azolla	0.00032	0.01319	0.9999	0.00022	0.01050443	0.999842
Ceratopteris	0.00042	0.01486	0.9999	0.00569	0.05184843	0.995725
Eichhornia	0.72026	0.34864	0.0862	0.00422	0.04059053	0.997307
Lemna	0.26941	0.21684	0.3658	0.00315	0.03728265	0.997945
Pistia	0.0934	0.13041	-0.071	0.00157	0.02466308	0.998841
Salvinia	1.33225	0.47458	0.0006	0.00083	0.02133753	0.9994
Spirodela	0.09711	0.13064	-2.447	0.00523	0.04458595	0.995885
Blank	0.01444	0.07031	0.9977	7.65438	1.14447321	-0.163197

For total coliforms, a wide variability in R^2 values was observed, ranging from -2.447 (*Spirodela*) to 0.9999 (*Azolla* and *Ceratopteris*). MSE and MAE values were consistently low for *Azolla* and *Ceratopteris* (MSE < 0.0005), while species such as *Eichhornia* and *Salvinia* showed the highest errors (MSE > 0.72).

For *E. coli*, six of the eight species exhibited R^2 values above 0.99, with generally low mean absolute errors (MAE < 0.05). *Azolla* presented the highest R^2 value (0.9998) and the lowest mean squared error (MSE = 0.00022), while the control showed the lowest R^2 value (-0.163) and the highest MSE (7.65).

B. Modeling with Penalty

Figure 3 presents the experimental and modeled coliform concentrations obtained by applying penalized removal coefficients *k* for the eight macrophyte species evaluated over the seven-day period. The models with environmental penalty reproduced the removal kinetics with slight discrepancies in most treatments, with *Ceratopteris* and the Blank control showing the highest concordance with the experimental data. It was observed that the penalty factors reduced the model's predictive capability for *Azolla* and *Spirodela*, while for *Eichhornia, Lemna*, and *Salvinia* the correspondence was maintained until days 4, 5, and 6, respectively, after which deviations in environmental parameters (pH, temperature, or TDS) increased the applied penalty.

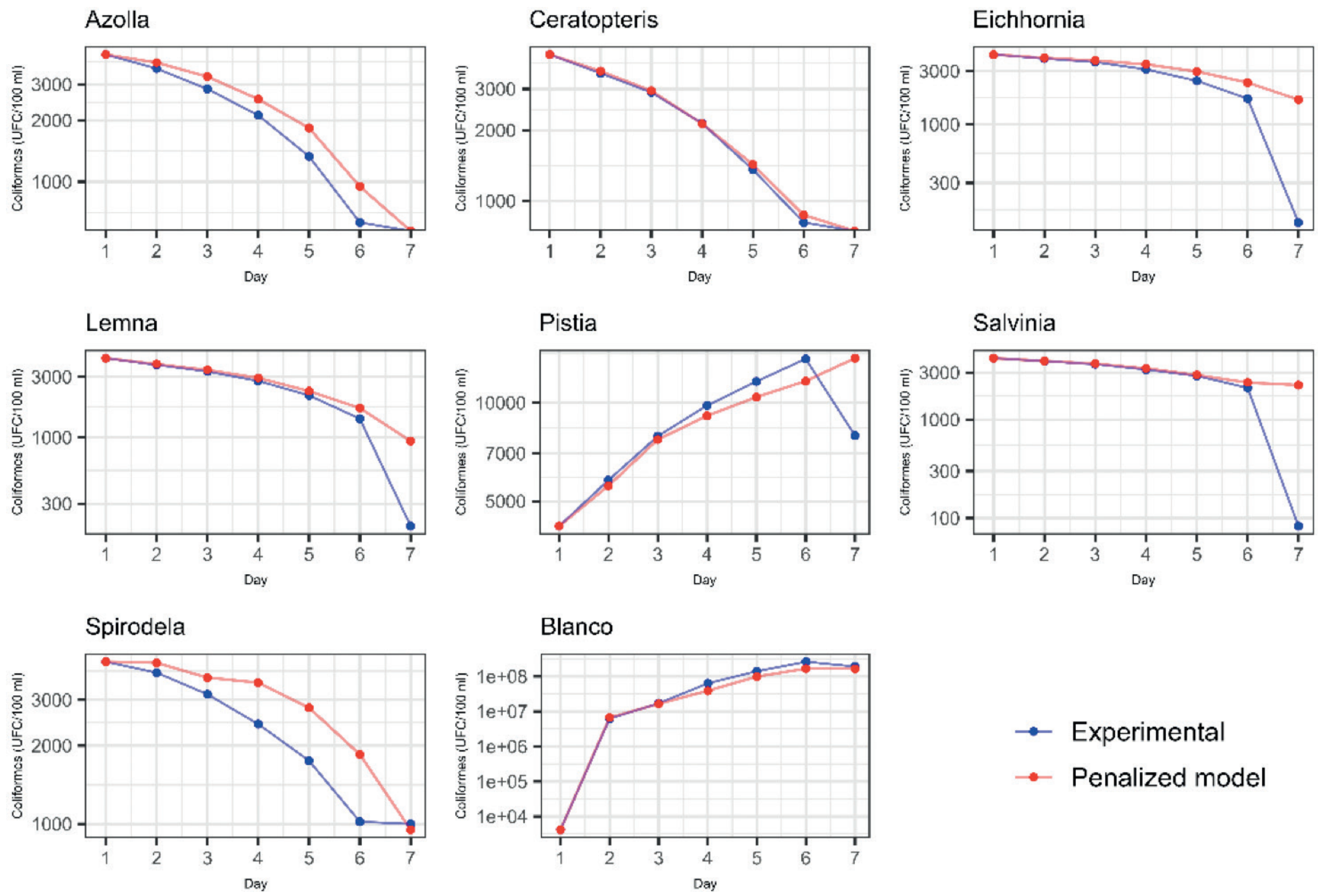

Fig. 3. Experimental versus Penalized-Modeled Removal Coefficients (*k*) for Total Coliforms.
Note: Error bars representing standard deviation (n=3) are present but appear imperceptible in the graphical representation due to the high magnitude of the initial concentrations and the scale of the Y-axis relative to the low operational variance observed among replicates.

Analysis of the modeling with environmental penalty identified that *Ceratopteris* showed the least overall effect, maintaining its physicochemical parameters within the 95% confidence interval throughout the trial. Conversely, *Azolla* and *Spirodela* showed the highest simultaneous deviations in the three evaluated environmental variables, resulting in the highest penalty factors.

Figure 4 presents the modeled *E. coli* concentrations with the application of a penalty factor based on the 95% confidence interval for the optimal parameters of pH, temperature, and (TDS), along with the experimental values. The penalized removal coefficients *k* used in the modeling show greater concordance with the experimental data for most treatments compared to the results observed for coliforms.

It was observed that the penalty factors had a lesser impact on the removal capacity for *Ceratopteris, Lemna, Pistia*, and *Salvinia*, while for *Azolla, Spirodela*, deviations in environmental parameters from the second day resulted in an increasing penalty that reduced the correspondence with the experimental data.

The validation metrics for the model with applied penalty are presented in Table II. This adjustment produced a more conservative representation of the removal process by reducing extreme k values and limiting overestimations.

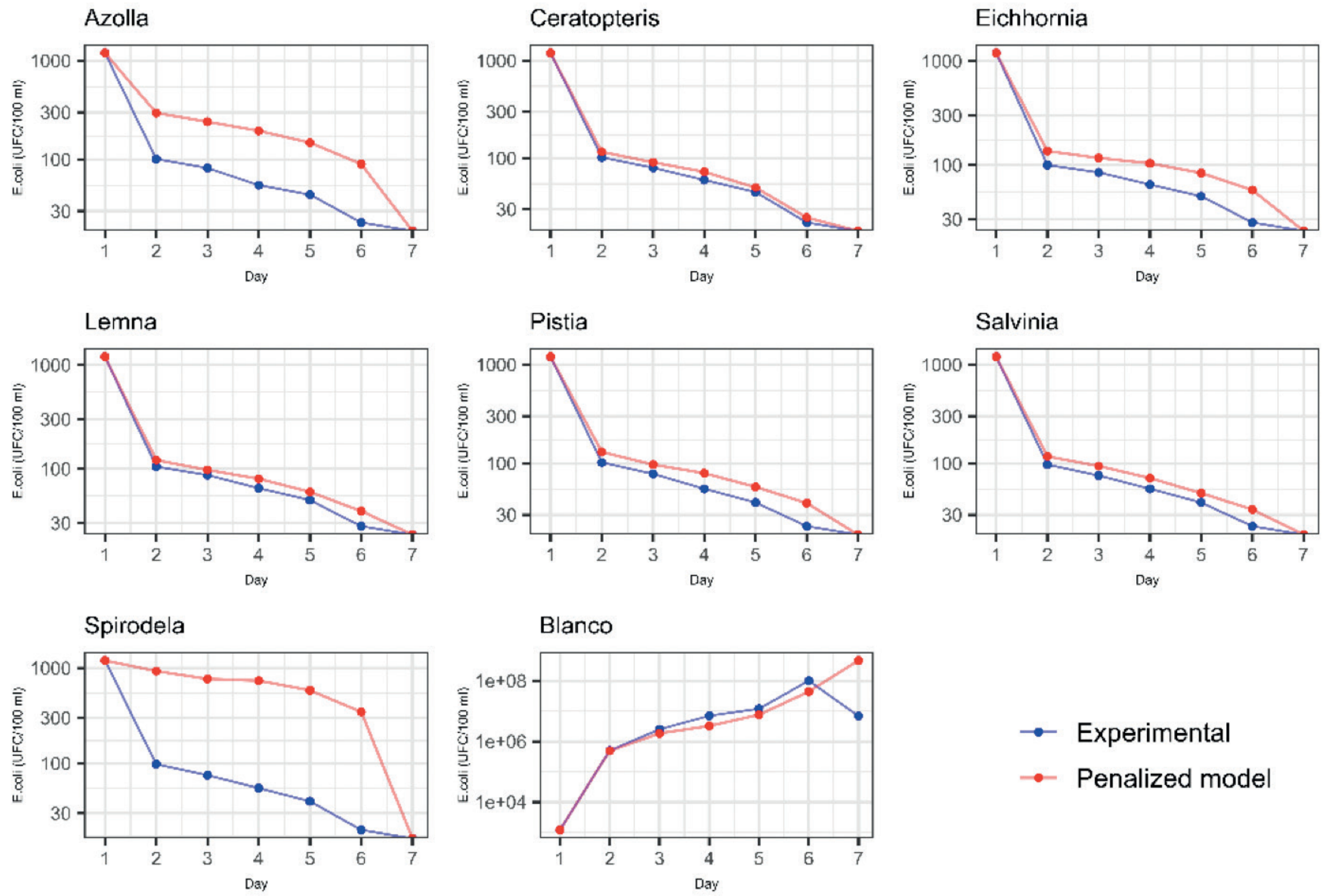

Fig. 4. Experimental versus Penalized-Modeled Removal Coefficients (k) for *E. coli*.
Note: Error bars representing standard deviation (n=3) are present but appear imperceptible in the graphical representation due to the high magnitude of the initial concentrations and the scale of the Y-axis relative to the low operational variance observed among replicates.

TABLE II
STATISTICAL VALIDATION OF PENALIZED MODELING (COLIFORMS AND *E. COLI*).

Plant	Coliforms			E. coli		
	MSE	MAE	R2	MSE	MAE	R2
Azolla	0.0339	0.1355	0.9934	0.5123	0.4723	0.6275
Ceratopteris	0.0016	0.0321	0.9997	0.007	0.0682	0.9947
Eichhornia	0.8102	0.4211	-0.0279	0.1129	0.2394	0.9279
Lemna	0.3053	0.257	0.2812	0.0282	0.1319	0.9816
Pistia	0.0829	0.1477	0.0490	0.0673	0.1895	0.9503
Salvinia	1.6779	0.5524	-0.2587	0.0373	0.1423	0.9732
Spirodela	0.0837	0.2068	-1.9723	2.2197	0.9512	-0.7475
Blank	0.0553	0.1889	0.9912	4.316	1.0823	0.3441

For total coliforms, the penalized model showed the highest fit in *Ceratopteris* (R^2 = 0.9997), with low error values (MSE = 0.0016, MAE = 0.0321), followed by Azolla (R^2 = 0.9934) and the Blank control (R^2 = 0.9912). In contrast, *Spirodela* presented the poorest fit (R^2 = -1.9723), followed by *Salvinia* (R^2 = -0.2587) and *Eichhornia* (R^2 = -0.0279), indicating that the penalized model did not adequately capture the removal dynamics in these treatments. Although *Pistia* showed a positive but low R^2 value (0.0490), its predictive performance remained limited, while *Lemna* showed only moderate fit (R^2 = 0.2812)

Unlike the pattern observed for total coliforms, the penalized model for *E. coli* showed the highest fit in *Ceratopteris* (R^2 = 0.9947), followed by *Lemna* (R^2 = 0.9816), *Salvinia* (R^2 = 0.9732), *Pistia* (R^2 = 0.9503), and *Eichhornia* (R^2 = 0.9279). In contrast, *Spirodela* presented the poorest fit (R^2 = -0.7475), while the Blank control showed limited predictive performance (R^2 = 0.3441). Although *Azolla* maintained a positive R^2 value (0.6275), its fit was lower than that observed in most macrophyte treatments. The highest error values were observed in *Spirodela* (MSE = 2.2197; MAE = 0.9512) and the Blank control (MSE = 4.316; MAE = 1.0823), indicating that the penalized model was less effective in representing highly variable or non-remediated systems.

Table III shows the comparison of the elimination coefficients k calculated for the coliform concentrations. The analysis shows that the environmental penalty greatly reduces the value of the penalized k coefficients compared to the modeled ones in most species, particularly towards the final days of the experiment. Drastic reductions are notable in

Azolla (day 7: modeled $k = 6.677 \rightarrow$ penalized $k = 0.252$) and *Ceratopteris* (day 7: 6.677 → 0.252), where the penalty attenuates the predicted removal rates. Anomalous cases such as *Eichhornia* show persistently negative values in the penalized k values, suggesting the penalized model is conservative in its removal dynamics.

TABLE III
EXPERIMENTAL AND MODELED REMOVAL COEFFICIENTS K FOR COLIFORMS.

Plant	Day	k_exp	k_mod	k_mod_pen	Plant	Day	k_exp	k_mod	k_mod_pen
Azolla	2	0.157	0.159	0.129	*Pistia*	2	-0.323	-0.316	0.081
	3	0.232	0.236	0.195		3	-0.309	-0.321	0.077
	4	0.295	0.29	0.244		4	-0.216	-0.213	0.145
	5	0.465	0.47	0.415		5	-0.169	-0.172	0.24
	6	0.746	0.758	0.62		6	-0.159	-0.163	0.36
	7	6.447	6.43	0.638		7	0.539	-0.209	0.448
Ceratopteris	2	0.182	0.188	0.058	*Salvinia*	2	0.069	0.073	0.182
	3	0.188	0.19	0.065		3	0.071	0.072	0.222
	4	0.304	0.297	0.084		4	0.125	0.127	0.17
	5	0.453	0.462	0.147		5	0.147	0.146	0.321
	6	0.518	0.529	0.231		6	0.281	0.277	0.822
	7	6.697	6.677	0.252		7	3.231	0.412	0.885
Eichhornia	2	0.082	0.084	-0.317	*Spirodela*	2	0.096	0.098	0.084
	3	0.072	0.071	-0.305		3	0.192	0.202	0.132
	4	0.15	0.154	-0.197		4	0.264	0.275	0.329
	5	0.235	0.232	-0.177		5	0.324	0.319	0.381
	6	0.366	0.37	-0.152		6	0.54	0.532	0.264
	7	2.548	0.543	-0.086		7	0.02	0.754	0.356
Lemna	2	0.121	0.12	0.121	Blank	2	-7.297	-7.025	-7.316
	3	0.12	0.112	0.123		3	-1.009	-1.01	-0.986
	4	0.172	0.167	0.158		4	-1.31	-1.342	-1.213
	5	0.262	0.266	0.287		5	-0.799	-0.804	-1.006
	6	0.424	0.427	0.44		6	-0.619	-0.604	-0.538
	7	1.946	0.642	0.492		7	0.311	0.164	-0.664

Note. *k_exp* = experimental removal coefficient; *k_mod* = modeled removal coefficient; *k_mod_pen* = modeled removal coefficient with environmental penalty applied.

Behaviors with high deviations were observed in species such as *Pistia*, where modeled k values that were negative (indicating theoretical bacterial growth) reverted to positive values after the penalty application (day 7: modeled $k = -0.209 \rightarrow$ penalized $k = 0.448$), aligning better with the experimental trend. The control (Blank) maintained negative values in all scenarios, confirming the absence of remediation capacity. The penalty introduces stability into the predictions, moderating extreme peaks such as those recorded in *Salvinia* (day 7: modeled k = 0.412 → penalized k = 0.885) and *Spirodela* (day 7: modeled k = 0.754 → penalized k = 0.356).

The removal coefficients for *E. coli* (table IV) show a generalized reduction in the penalized removal coefficients k compared to the modeled coefficients without penalty in the final days, like the effect observed for coliforms. This is evidenced on day 7, where the environmental penalty drastically attenuates the predicted removal rates. Furthermore, notable decreases are observed in *Azolla* (modeled k = 3.127 → penalized k = 0.62), *Ceratopteris* (modeled k = 3.011 → penalized k = 0.518), and *Eichhornia* (modeled k = 3.389 → penalized k = 0.67), where the penalized values represent only 16-20% of the modeled values without penalty. This trend is consistent across all macrophytes, indicating that the penalized model generates more conservative predictions.

TABLE IV
EXPERIMENTAL AND MODELED REMOVAL COEFFICIENTS K FOR *E. COLI*

Plant	Day	k_exp	k_mod	k_mod_pen	Plant	Day	k_exp	k_mod	k_mod_pen
Azolla	2	2.475	2.469	2.082	*Pistia*	2	2.465	2.398	2.421
	3	0.208	0.21	0.214		3	0.268	0.26	0.188
	4	0.399	0.413	0.24		4	0.349	0.352	0.209
	5	0.223	0.228	0.325		5	0.319	0.304	0.336
	6	0.649	0.643	0.469		6	0.553	0.546	0.567
	7	3.136	3.127	0.62		7	3.136	3.229	0.69
Ceratopteris	2	2.465	2.513	2.128	*Salvinia*	2	2.515	2.483	2.422
	3	0.243	0.248	0.238		3	0.257	0.258	0.289
	4	0.288	0.281	0.224		4	0.31	0.31	0.204
	5	0.288	0.294	0.285		5	0.319	0.326	0.352
	6	0.716	0.745	0.486		6	0.553	0.557	0.685
	7	3.091	3.011	0.518		7	3.136	3.157	0.751
Eichhornia	2	2.485	2.465	2.573	*Spirodela*	2	2.505	2.643	2.18
	3	0.163	0.153	0.279		3	0.268	0.273	0.244
	4	0.268	0.267	0.285		4	0.31	0.312	0.241
	5	0.262	0.266	0.359		5	0.319	0.319	0.323
	6	0.58	0.55	0.545		6	0.693	0.7	0.589
	7	3.332	3.389	0.67		7	2.996	2.843	0.585
Lemna	2	2.436	2.485	2.485	Blank	2	-6.032	-6.121	-6.275
	3	0.188	0.176	0.194		3	-1.609	-1.583	-1.682
	4	0.292	0.296	0.22		4	-1.03	-1.004	-1.047
	5	0.262	0.258	0.337		5	-0.539	-0.531	-0.466
	6	0.58	0.591	0.564		6	-2.12	-2.163	0.145
	7	3.332	3.285	0.639		7	2.674	-4.381	0.714

Note. *k_exp* = experimental removal coefficient; *k_mod* = modeled removal coefficient; *k_mod_pen* = modeled removal coefficient with environmental penalty applied.

Finally, anomalous behavior was observed in the control (Blank), where the penalized k values changed from negative to positive on days 6 and 7 (modeled k = -2.163 → penalized k = 0.145 and modeled k = -4.381 → penalized k = 0.714), contradicting the experimental expectation of continuous bacterial growth. Cases such as *Spirodela* showed less impact from the penalty, maintaining relatively stable values across all days. The smoother transition between consecutive days in the penalized k values suggests that the analysis criterion mitigates abrupt variations in the predictions, which is particularly evident in the stabilization of the day 7 values across all species.

IV. DISCUSSION

The prediction of mathematical models under real conditions constantly faces the inherent sensitivity of complex interactions in phytoremediation systems [19]. The environmental penalty model developed in this study proves to be a support tool for incorporating the influence of environmental variables on the removal kinetics of coliforms and *E. coli* [20]. Unlike models based solely on time, the inclusion of parameters such as pH, temperature, and (TDS) allowed for dynamic adjustment of the removal coefficient k, reflecting the biological sensitivity to medium conditions.

According to the classification of design tools proposed by [21], the model developed in this study represents an evolution beyond traditional approaches. While traditional “Rules of Thumb” and simple regression equations show massive output variability often spanning several orders of magnitude our model utilizes calibrated biological penalties to provide standardized predictions. Furthermore, while the widely used first-order models assume a constant removal rate, our approach addresses the “parameter uncertainty” highlighted by [21] by making *k* dynamic; and compared to complex mechanistic models that are often “less manageable” for designers due to excessive parameter requirements [21], our model offers a practical “middle ground” that maintains the simplicity of first-order kinetics while adding the realism of environmental stressors.

This operational simplification is particularly advantageous when compared to mechanistic-kinetic models such as that of [9], who proposed that the overall removal rate is the sum of individual coefficients for sedimentation, filtration, and mortality $K = k_s + k_f + k_m$ While the model in [9] offers high detail for tropical conditions, it requires the determination of complex physical parameters—such as particle settling velocity and filter pore size that are difficult to measure in situ. In contrast, our environmental penalty approach captures the sensitivity of these “mortality” and “adsorption” processes through easily monitorable physicochemical variables (pH, temperature, and TDS), facilitating its use as a real-time support tool without sacrificing biological rigor.

The selection of pH and temperature as primary penalty factors is supported by the biochemical principles outlined by [8], who emphasize that these variables directly dictate microbial enzymatic activity and the solubility of contaminants. By incorporating these stressors, the model aligns with the ‘interplay of disciplines’ necessary for modern bioremediation, where engineering design must account for the physiological limits of the decontaminating microbiota [8].

This approach is particularly relevant given that kinetic parameters, such as absorption rates, exhibit high variability when extrapolated beyond the original calibration conditions [22]. Therefore, models incorporating parametric adjustment mechanisms, such as environmental penalty, are more suitable for addressing the inherent complexity of phytoremediation systems in real scenarios [23].

Among the models with environmental penalty, it was demonstrated that physicochemical factors affected removal efficiency, with the greatest impacts observed in *Azolla*, *Pistia*, and *Spirodela* for coliforms, and in *Azolla* and *Spirodela* for *E. coli*. This difference in penalty can be attributed to two factors: on one hand, *Azolla*, *Pistia*, and *Spirodela* critically depend on mechanisms sensitive to environmental variations, such as physical filtration through extensive root systems and the release of antimicrobial exudates [1]. These processes are highly vulnerable to fluctuations in parameters like pH and dissolved solids concentration, which explains the higher penalties recorded. On the other hand, as it was a system without water renewal, the environmental parameters (pH, TDS, temperature) varied, as a result of the metabolic activity of the plants themselves. Species like *Azolla* and *Pistia* actively alter their microenvironment through the exudation of organic compounds and modification of oxygenation [6], which generated conditions outside the optimal range established in the first days of the trial. This phenomenon amplified the penalty in species with a greater capacity for environmental modification, while plants like *Ceratopteris* and *Lemna* maintained more stable conditions, reducing their penalty.

Species such as *Ceratopteris* and *Lemna* showed less susceptibility to environmental variations, possibly due to more dynamic remediation mechanisms or physiological adaptations that confer greater tolerance and stability over time [16]. The greater penalty observed for *E. coli* compared to total coliforms, particularly in *Azolla* and *Spirodela*, further suggests that this bacterium presents sensitivity to environmental variations, possibly related to more specific metabolic requirements [18].

The penalty mechanism, in this sense, helps to avoid overestimations compared to simple models, especially on final days where experimental *k* values tend to be anomalous. For example, in *Azolla* sp., the modeled k value for coliforms on day 7 was 6.43, while the penalized version reduced this value to 0.638, bringing it closer to the overall trend observed throughout the timeline for each plant. Similarly, in *Pistia stratiotes*, the penalized model reversed biologically implausible negative modeled *k* values (e.g., –0.209) to positive values (0.448), aligning better with experimental behavior [17].

Regarding *C. thalictroides*, the modeled *k* value of 6.677 on day 7 while high compared to previous days reflects a theoretical peak derived from the polynomial extrapolation of the growth and removal phase. However, the application of the environmental penalty effectively moderates this coefficient, ensuring that the final prediction accounts for the physiological stress observed in the experimental system.

However, the model still presented limitations in species such as *E. crassipes* and *S. auriculata*, where the penalized *k* values were persistently negative or showed little correspondence with the experimental data towards the end of the trial. This would suggest the presence of late removal mechanisms, such as the release of secondary metabolites or changes in biofilm structure, not captured by the considered environmental variables [30].

Several key advantages and limitations are identified. Among the primary advantages, the model demonstrates a significant capacity for the correction of biological implausibility; as highlighted in literature, simple linear fits often produce “anomalous” data, yet this model successfully corrected negative *k* values in species like *P. stratiotes*, converting them into positive, realistic coefficients [17].

Furthermore, the model offers dynamic predictive power by making the removal coefficient sensitive to pH and TDS, thereby avoiding the overestimations common in static models. This sensitivity is scientifically grounded in the biochemical principles described by [8], which establish that pH and temperature directly dictate microbial enzymatic induction and the solubility of contaminants, consequently, the model functions as a conservative design tool and a sustainable alternative to traditional disposal methods, providing a critical “safety margin” for wastewater reuse planning in agricultural scenarios.

However, certain limitations must be acknowledged to contextualize these findings. First, the model relies on hydraulic idealization; consistent with constraints mentioned by [21], it assumes ideal flow conditions, whereas field applications often encounter “short-circuiting” and “dead zones” not captured by this batch-data-based version. Second, the model exhibits species-dependent thresholds, meaning its accuracy depends heavily on the prior characterization of critical limits for each species, which may vary across different geographical and operational contexts.

Additionally, a limitation identified in the broader context of bioremediation [8] is the potential generation of toxic metabolic byproducts; our current model focuses on the removal of primary indicators *E. coli* and coliforms but does not yet simulate secondary metabolites.

There is a distinct focus on the water column, currently omitting the complex biofilm dynamics on the root surface. As [21] notes that purification is largely a biofilm-mediated process, and [9] demonstrates that adsorption in the rhizosphere is a major removal pathway, future iterations must integrate these interactions to improve the predictive fit in species such as *E. crassipes* and *S. auriculata*.

Statistical validation confirmed that the model with environmental penalty improved predictive capability in several cases, although performance varied among species. For total coliforms, the best fits were observed in *C. thalictroides* and *Azolla sp.*, while lower fits were mainly observed in *Spirodela sp., S. auriculata*, and *E. crassipes*. For E. coli, the lowest fits occurred in *Spirodela sp*. and the Blank control, indicating that the penalty was not sufficient to compensate for high variability or unmodeled factors in some treatments [3].

Specifically, the negative R2 values reported for *Spirodela sp.* and blank control indicate that the intrinsic variability of the experimental data in these cases does not follow the proposed first-order linear trend. Statistically, this implies that the model is a less efficient predictor than the simple mean of the observations. This phenomenon may be attributed to erratic removal behaviors or the influence of non-linear biological factors, such as the release of metabolites or the maturation of biofilms, which are not currently integrated into the model.

These results aim to shed light on the incorporation of environmental variables in phytoremediation, especially in open systems or with natural fluctuations [24]. Nevertheless, they also highlight the need to include additional biological factors (such as biofilm dynamics, microbial competition, or the production of specific exudates) to improve predictive accuracy in species with complex removal mechanisms [26]. The environmental penalty approach constitutes a step towards more realistic models, although its applicability still depends on the prior characterization of critical thresholds for each species and operational context.

From a practical standpoint, the modeled phytoremediation system presents a low-cost alternative for rural communities where conventional treatment plants are economically unfeasible. By using the environmental penalty model, designers can optimize species selection without the need for expensive real-time microbial monitoring, relying instead on accessible physicochemical sensors. While this study utilized batch-scale data, the model provides a scalable framework for constructed wetlands, although future transitions to large-scale systems must account for hydraulic residence time and competitive microbial dynamics to maintain cost-efficiency.

V. CONCLUSION

The study successfully developed a first-order mathematical model with nuances that define its applicability. The kinetic model with dynamic environmental penalty for pH, temperature, and TDS demonstrated its ability to predictively adjust the removal coefficient (k) based on environmental conditions. Statistical validation showed that the model adequately represented bacterial removal trends in several treatments, particularly for *E. coli*. The environmental penalty allowed the influence of pH, temperature, and TDS on the removal coefficient to be incorporated, producing more conservative predictions by reducing extreme k values and limiting overestimations. For total coliforms, the penalized model showed the best fit in *C. thalictroides* (R^2 = 0.9997) and *Azolla sp.* (R^2 = 0.9934), whereas *Spirodela sp.*, *S. auriculata*, and *E. crassipes* showed poor performance. For *E. coli*, the best fits were observed in *C. thalictroides, L. minor, S. auriculata, P. stratiotes*, and *E. crassipes*, although the model showed lower performance in *Spirodela sp.* and the Blank control. However, the model has specific limitations. It showed variable sensitivity among species, with suboptimal performance in *Spirodela sp*. (R^2 = -0.7475) and poor fits in blank control (R^2 = 0.3441), indicating that the environmental penalty does not fully compensate for the high variability in systems without macrophytes or in species with complex removal mechanisms. Similarly, predictive effectiveness was significantly lower for total coliforms compared to *E. coli*, particularly in species such as *E. crassipes* and *S. auriculata* where negative penalized k values persisted.

The study demonstrates that a penalized model is less simplistic than a time-based one; from a practical standpoint, this model serves as a predictive tool for the design and operation of decentralized wastewater treatment systems, such as constructed wetlands in rural or peri-urban areas. By accounting for fluctuating environmental parameters, stakeholders can use the model to simulate disinfection efficiency under varying climatic conditions, allowing for the selection of the most resilient species e.g., *C. thalictroides or L. minor* to ensure water safety for agricultural reuse or safe discharge.

However, for real-world scenarios, it is necessary to consider the species-dependent specificity of removal mechanisms and the potential interference of non-linear late-stage microbial processes. To enhance the model's practical relevance for field applications, several strategic improvements are suggested to bridge the gap between experimental data and environmental engineering. Future iterations should transition from static (batch) data to continuous flow scenarios, allowing the model to account for convective transport and hydraulic loading rates typical of functional treatment systems.

Furthermore, future research should integrate additional biological variables—such as biofilm dynamics, dissolved oxygen, and the density of both live and senescent biomass to improve predictability in species with complex kinetic behaviors. Refining physical parameters to include root surface area (rhizosphere) and the diversity of attached microbiota would help the model account for the complex removal mechanisms that standard first-order kinetics may overlook. Finally, there is a clear need to calibrate specific penalty thresholds for each operational context; this would allow the model to better reflect interactions such as microbial competition and the release of secondary metabolites, ensuring its reliability as a decision-support tool in diverse environmental settings.

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The authors declare that no artificial intelligence tools were used.

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